# Brenna Ren

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## EDUCATION

**Yale University** — New Haven, CT May 2030
*Bachelor of Science, Computer Science and Applied Math*

**The Harker School —** San Jose, CA May 2026
*GPA: 4.00/4.00 (UW), 4.65/4.80 (W) • SAT: 1570 (770 RW, 800 M)*

- Relevant Coursework: Data Structures, Neural Networks, Computer Architecture, Robot Kinematics Software, Compilers & Interpreters, Linear Algebra, Discrete Math, Multivariable Calculus, Differential Equations

## EXPERIENCE

**FRC Team 1072: Harker Robotics** August 2022–May 2026
*Executive President (25–26), Technical Vice President (24–25), Software Director (23–24)*

- Achieved 2x World Championship Qualification and 2x Offseason Champions by leading **end-to-end robot development** (design, fabrication, programming, and testing) across a **102-member team with a $50K budget**
- **Grew membership by 15%** and led 100+ annual meetings, enabling hands-on involvement for all members
- Earned Dean's List Semi-Finalist recognition for exhibiting outstanding leadership capabilities

**Garcia Summer Research Scholar - Stony Brook University** June 2024–March 2026

- Published **first-author manuscript in MRS Communications** by developing an **ML model** that quantified polydispersity effects on polystyrene thin-film thickness for informing polystyrene coating industrial applications
- Collected 300+ experimental samples and analyzed with **Python** and statistical modeling
- Presented research at MRS, ACS, and APS conferences

## PROJECTS

**2023-26 FRC 1072 Robot Code** August 2022–Present

- Maintained a **20K LOC Java control codebase** with 150+ personal Git commits
- Reduced cycle times by ~20% by implementing driver-assist auto-alignment and **autonomous routines** on top of tuned PID and motion-profiling **closed-loop controllers**
- Achieved **accurate real-time pose estimation** by implementing a Kalman filter fusing odometry, gyro, and vision data, enabling reliable autonomous navigation

**Deep Learning for Medical Signal Analysis** August 2024–June 2025

- Improved RBD diagnostic **accuracy from 80% to 97%** by designing an unsupervised deep learning model in **Tensorflow/Keras** to detect the condition from ECG data
- Presented at the 23rd IEEE/ACIS SERA conference as first author, **published in Springer SCIS (top 15 papers)**

**Quantum Key Distribution Protocol Simulation** June 2023–April 2024

- Validated theoretical QKD security by building **Python Qiskit** simulations comparing QBER across protocols in ideal and noisy environments using IBM Quantum Platform

## HONORS

- USACO Gold Division • AIME Qualifier • NCWIT Aspirations in Computing (2x National Honorable Mention) • ACSL Senior Division Bronze Medal (3x Finals Qualifier)

## SKILLS

- **Programming:** Java, C++, Python, Git, Linux, TensorFlow, Keras
- **Languages:** English, Mandarin Chinese